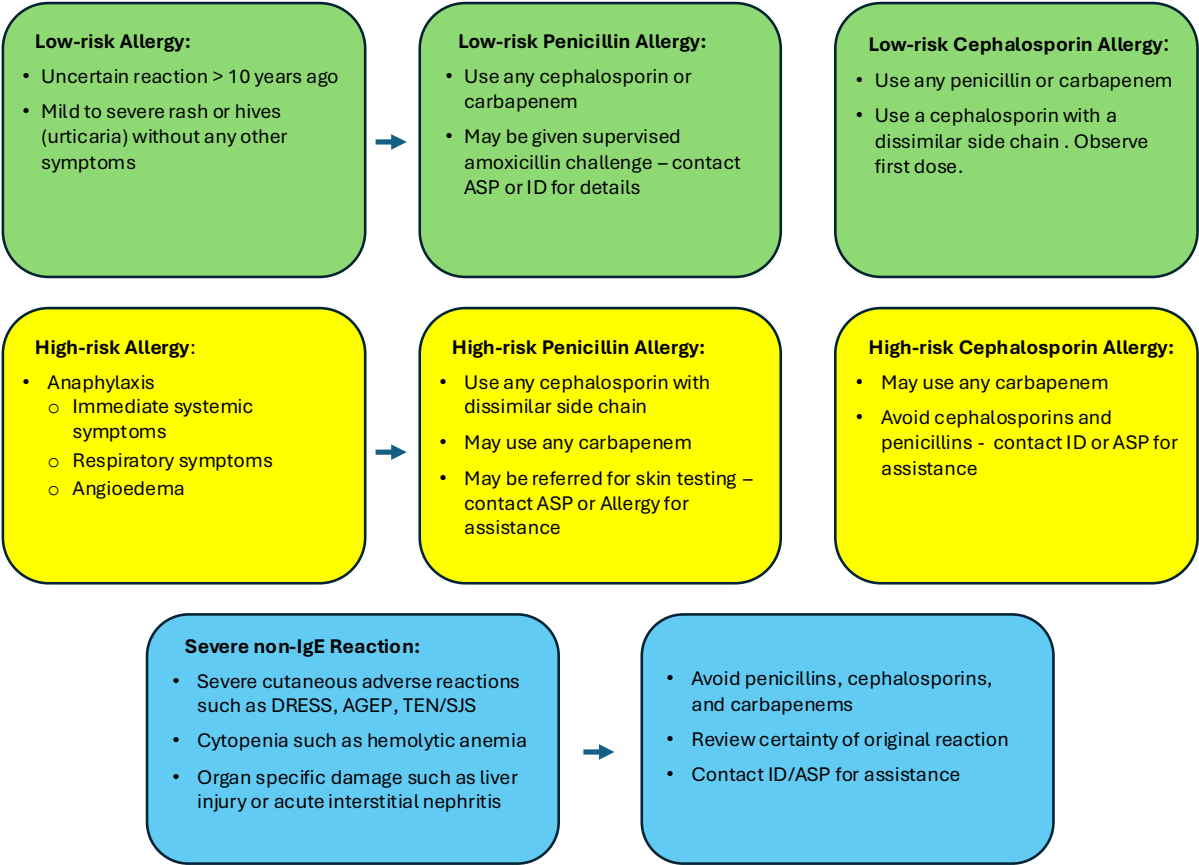


Beta-Lactam Reactions
Assessment of Reaction & Selection of Alternative Beta-Lactams

- Penicillin allergies wane over time
- Cephalosporins are preferred over non-beta-lactams for safety and effectiveness.
- Beta lactam cross-reactivity is driven by similar side chains.
- **Patients with a history of anaphylactic reaction to penicillin can receive a cephalosporin with non-related side chains, such as cefazolin, without further testing or precaution.**
- Use **Figure 1** for assessment of allergy. See **Figure 2**, page 2, for shared side chains.

Figure 1. Assessment of Beta-lactam Allergy



Beta-lactam Antibiotic Cross-reactivity Chart

- Cross-reactivity charts should **only** be consulted after assessing severity of reaction.
- Consult ID or ASP for assistance in assessing specific cases

Figure 2. Beta-lactam Antibiotic Cross-reactivity Chart

	Cefazolin	Cefaclor	Cefadroxil	Cefepime	Cefotaxime	Cefoxitin	Cefprozil	Ceftazidime	Ceftriaxone	Cephalexin	Amoxicillin	Ampicillin	Aztreonam	Cefiderocol	Penicillin*
Cefazolin															
Cefaclor										X		X			X
Cefadroxil							X			X	X				X
Cefepime					X										
Cefotaxime				X					X						
Cefoxitin															
Cefprozil			X								X				X
Ceftazidime													X	X	
Ceftriaxone				X	X										
Cephalexin		X	X									X			X
Amoxicillin			X				X								X
Ampicillin		X								X					X
Aztreonam								X						X	
Cefiderocol													X		
Penicillin*		X	X				X			X	X	X			
Khan et al 2022, *Reddy et al 2026, adapted. Contact ID/ASP for guidance															

References

- [Khan et al, Drug allergy: A 2022 practice parameter update J Allergy Clin Immunol, 2022, 150\(6\)133](#)
- [Chaudhry et al., Cephalosporins: A Focus on Side Chains and β-Lactam Cross-Reactivity. Pharmacy, 2019, 7\(3\)103](#)
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- [Reddy et al, Intracameral Cefuroxime Use in Cataract Surgery After Penicillin Allergy Reclassification JAMA Ophthalmol. 2026, 2026, 144\(4\):325](#)